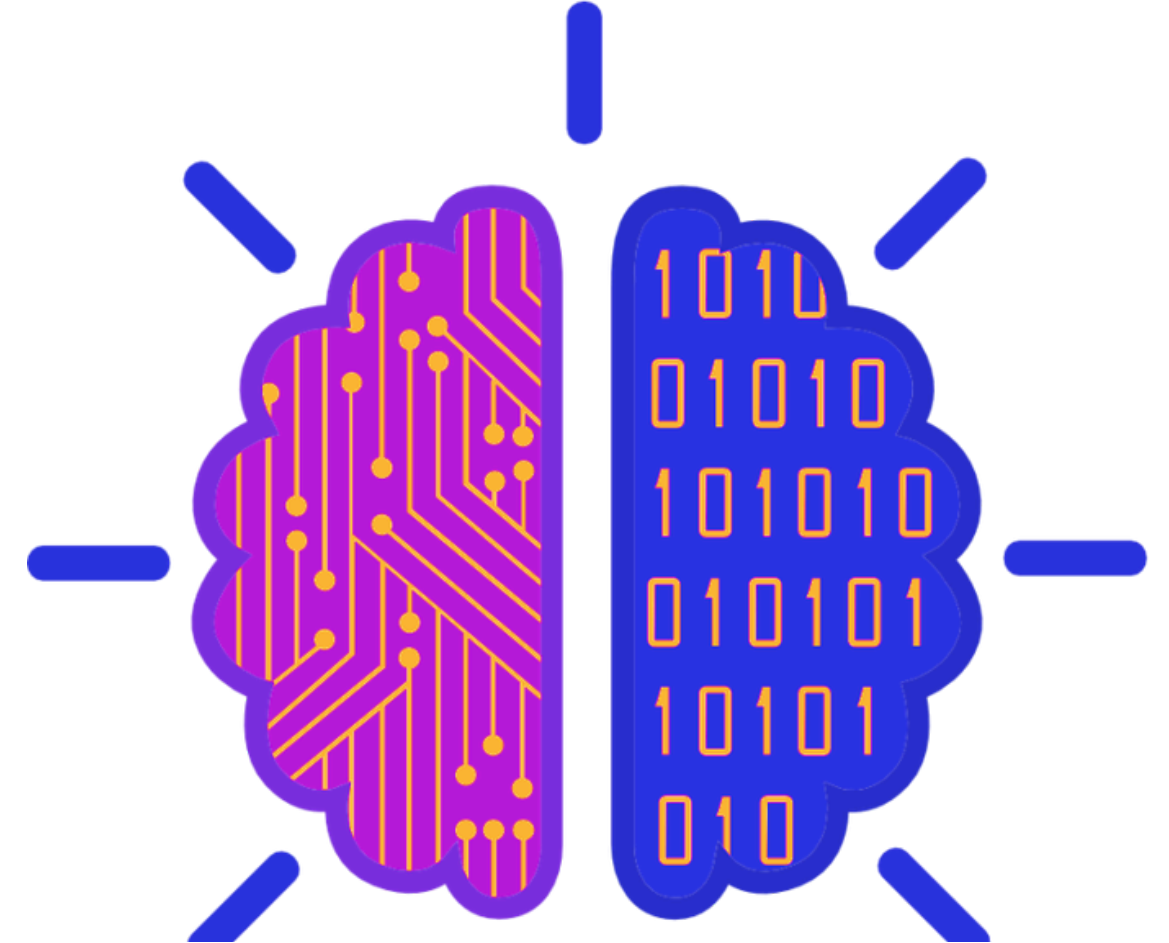


MAKEATHON 3.0

II YEAR

- **Problem Statement ID – SIH25229**
- **Problem Statement Title - Identification of URL Based Attacks from IP Data**
- **Theme- Cybersecurity**
- **PS Category- Software**
- **Team Name - The Zenithal**



Identification of URL Based Attacks from IP Data

Problems Faced

- Users receive phishing links through **WhatsApp and SMS** disguised as bank alerts, delivery updates, or payment notifications.
- Shortened **URLs hide the real destination**, making malicious links appear valid.
- Most users cannot easily distinguish between safe and phishing links **before clicking** them.
- Traditional security systems **fail to detect** newly created phishing domains and zero-day attacks.
- Clicking malicious links **can result** in stolen credentials, financial fraud, or malware infection.

Solution Approach

- **Automatically scans suspicious links** from WhatsApp, SMS, and web browsing before users open them.
- Analyzes URL structure, domain details, and IP infrastructure to **uncover hidden phishing websites**.
- **Uses AI models** to detect disguised or newly created malicious links in real time.
- Provides **instant security alerts** when a suspicious or dangerous link is detected.
- Shows a clear explanation so users understand **why the link is unsafe**.

Uniqueness



Messaging-aware security – scans URLs shared through WhatsApp and SMS and other platforms, which most traditional systems do not monitor.



Multi-layer AI detection – combines ML, Cyber SLMs, and reasoning models for deeper threat analysis.



IP-domain correlation analysis – detects suspicious websites by comparing domain identity with hosting IP infrastructure.



Explainable AI decisions – provides clear reasons why a URL is marked safe, suspicious, or malicious.

Objectives



Detect malicious and phishing URLs in real time before users open them.



Use AI models to improve detection accuracy beyond traditional blacklist systems.



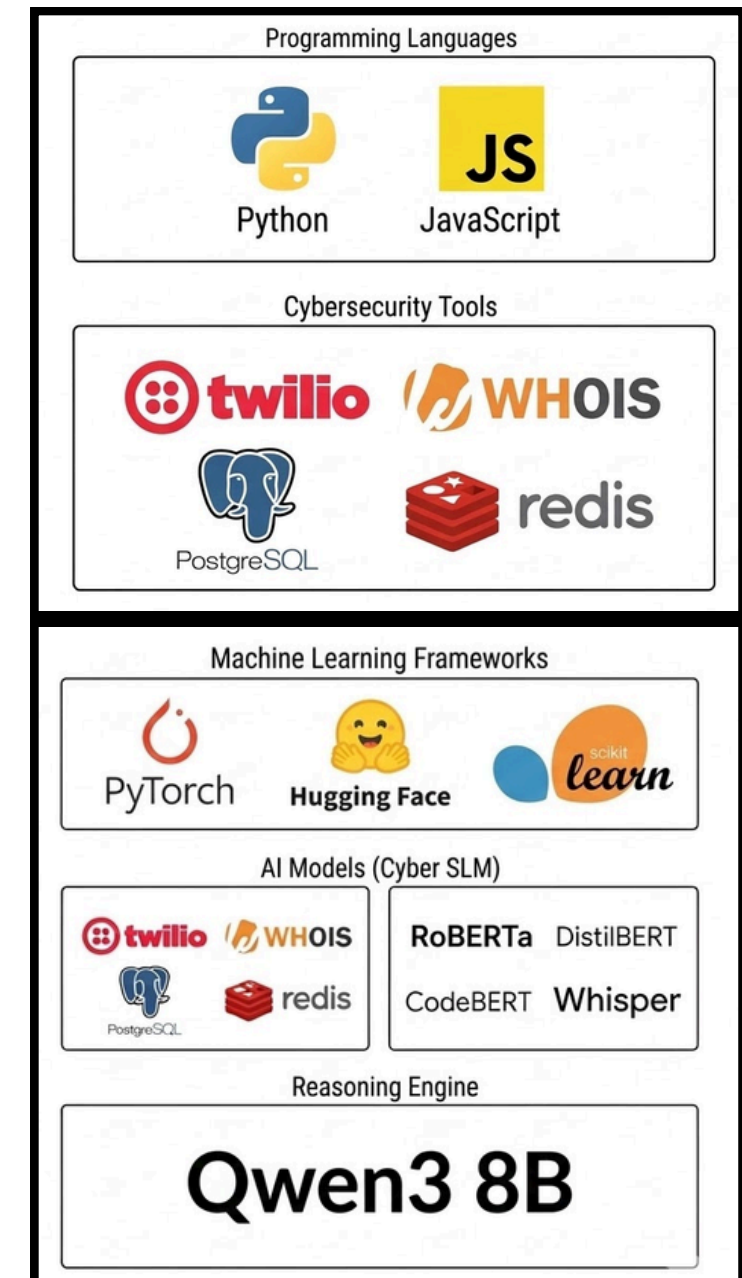
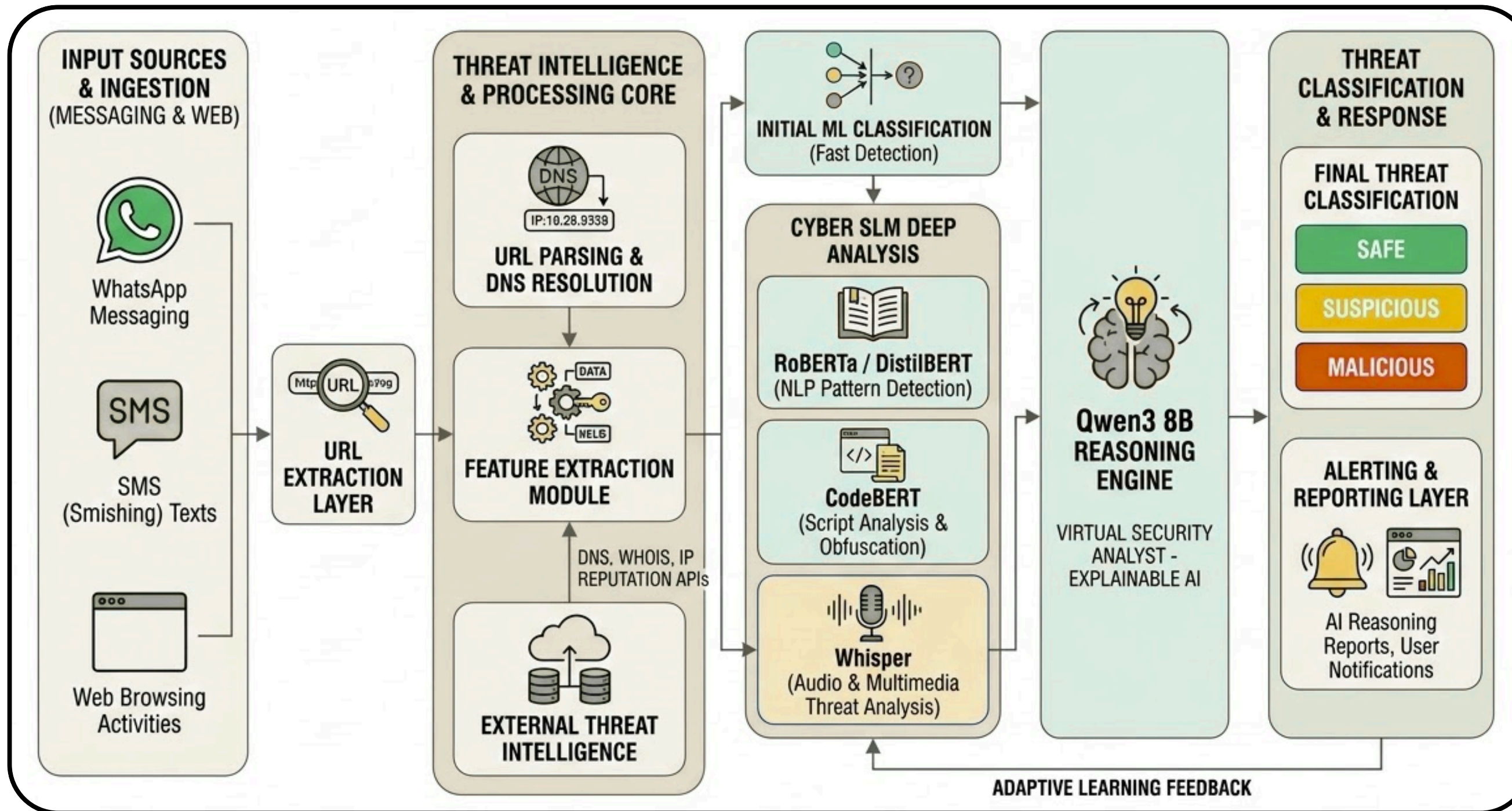
Analyze URL structure, domain information, and IP data to identify hidden cyber threats.



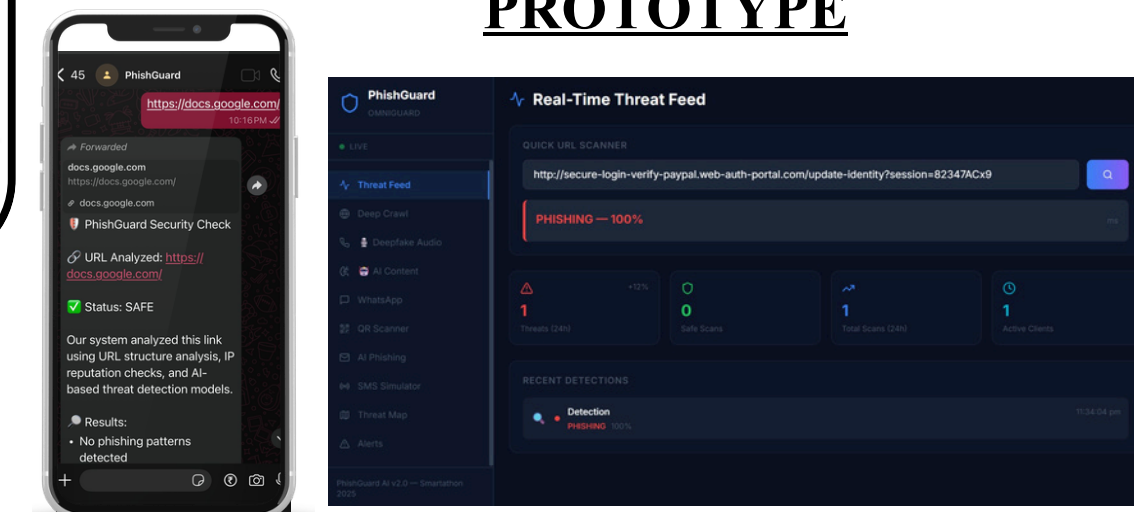
Protect users from phishing, smishing, and malware links shared through web or messaging platforms.



ARCHITECTURE DIAGRAM



PROTOTYPE



FEASIBILITY

Technical Feasibility

The system can be built using existing technologies such as machine learning models, DNS/IP intelligence APIs, and open-source cybersecurity tools.

Operational Feasibility

The solution runs automatically in the background and scans URLs from WhatsApp and SMS without affecting normal user activities.

Implementation Feasibility

Python, PyTorch, Hugging Face, and React provide a seamless, integrated stack for AI model deployment, backend logic, and interactive dashboards.

Data Feasibility

Public phishing datasets and threat intelligence sources provide sufficient data to train and improve the detection models.

VIABILITY

Economic Viability:

Using open-source AI models and cloud infrastructure significantly reduces development and operational costs.

Market Viability:

The growing number of phishing and smishing attacks creates a strong demand for intelligent URL threat detection solutions.

Scalability

The architecture can scale to analyze thousands or millions of URLs in real time for enterprise-level deployment.

Long-Term Viability:

Adaptive learning enables the system to evolve with new cyberattack techniques, ensuring long-term effectiveness.

IMPACTS AND BENEFITS

Impact on Users

- Protects individuals from **phishing, smishing, and malicious links**.
- Prevents **financial fraud and identity theft** through early threat detection.
- Provides clear **security alerts before** users interact with suspicious URLs.

Impact on Society

- Reduces the **spread of cyber fraud and online scams**.
- Improves **public awareness** and safety in digital communication.
- Builds greater **trust in internet services** and online platforms.

Impact on Government

- Supports **national cybersecurity** monitoring
- Helps detect **large-scale phishing** and cybercrime campaigns.
- Strengthens national digital security and **cyber threat intelligence**.

Impact on Organizations

- Protects **employees from phishing-based** cyber attacks.
- Reduces **financial and data loss** caused by malicious links.
- Enhances **enterprise cybersecurity** with proactive AI threat detection.

Benefits

- **Early threat detection** – identifies malicious links before users open them.
- **Safer messaging experience** – protects links shared through WhatsApp and SMS.
- **AI-based smart analysis** – detects new and unknown phishing attacks.
- **Clear security explanation** – tells users why a link is dangerous.
- **Reduced cyber fraud** – helps prevent scams and financial loss.
- **Faster threat investigation** – helps cybersecurity teams analyze attacks quickly.
- **Scalable security solution** – capable of monitoring large volumes of URLs.

Challenges Faced

Phishing and malicious URL attacks are among the fastest-growing cyber threats globally, with over 1 billion phishing attacks recorded in 2023 alone. Existing detection systems lack real-time multi-channel coverage across WhatsApp, SMS, and web traffic, leaving users vulnerable to socially engineered threats.

Reference Problem

- In 2023, a large-scale WhatsApp phishing campaign in India targeted thousands of users with malicious URLs disguised as government links, resulting in financial fraud and data theft exposing critical gaps in mobile messaging threat detection.
<https://shorturl.at/example1>
- Transformer-based models like RoBERTa and DistilBERT have demonstrated over 98% accuracy in phishing URL detection, yet remain largely undeployed in real-time mobile threat systems.
<https://doi.org/10.48550/arXiv.2311.04913>
- CodeBERT has proven effective in detecting encoded malicious payloads and obfuscated query structures embedded in URLs, a technique increasingly used by attackers to bypass traditional filters.
<https://doi.org/10.18653/v1/2020.findings-emnlp.139>

SDG'S Aligned

